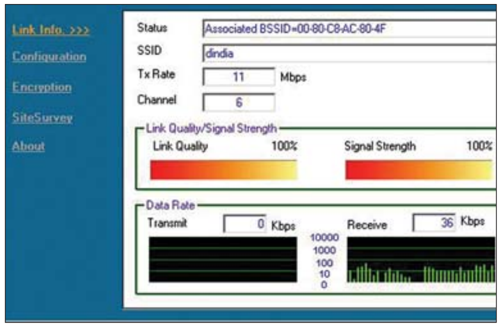

Place the Access Point such that the users get the optimal signal reception.

SSID and encryption key, are automatically picked up by the adapter card from the Access Point.

STEP 3 Check the signal strength available at the place the wireless desktop, or laptop is placed by double clicking the utility in the taskbar.

While setting up a wireless network, it is important to remember to place the access point in a proper place: metal tables, cupboards, and cabinets filled with clothes can cause a drop in throughput. Though a wireless connection is the best choice for a multi-storey house, there will be obstacles such as walls, flooring, metal tables and microwave ovens which might lower the data throughput. To avoid this, move the access point around to find the place where the clients get the maximum signal strength. The same process will have to be repeated for the clients' placement.

Tips And Tricks

Clear to Fly

Window panels with a metallic coating, wooden stuff, etc. greatly contribute to signal disturbance, thereby resulting in connection drop outs. Careful planning is required while placing the Access Point. Sometimes, one needs to set up multiple Access Points.

Location of the Access Point (AP)

The location of the AP is of utmost importance. It depends upon parameters such as line-of-sight and height from floor level. To get broader coverage per AP, you need to make sure the AP is in the line-of-sight from many different directions at the same time. Similarly, the height at which the AP is placed can influence the coverage area. Depending upon the internal structure, the range will increase or decrease as you move the AP up or down.

Reflective/Absorbing Materials

Metal causes multipath issues due to reflection, and it could impede signals. The signal deterioration will depend on how well the container is sealed, and on the thickness of the metal. 802.11g has lesser problems because of OFDM (orthogonal frequency division multiplexing). Still, a given access point won't provide the same range where there's a lot of metal, than it would in a typical office environment.

Frequency Band Issues

The band also plays an important role. The lower the band, the more tenacious it is (2.4 GHz vs. 5 GHz). Therefore, 802.11b or g is more reliable than 802.11a.

Security

Without stringent security in place, installing a WLAN is the equivalent of putting up Ethernet ports everywhere. Even if you use the default WEP security, it's not safe enough due to its dependence on a static encryption key. WPA (Wi-Fi Protected Access) is more preferred. Further, inline WPA2 with still better security is also coming up.

Government Regulations

As far as IEEE networking standards are concerned, the Indian government Wireless Planning Commission (WPC) has permitted unlicensed usage of the IEEE 802.11b wireless networking protocol that allows for WiFi connectivity at 11 Mbps. The same is not the case with the IEEE 802.11g standard: the regulations are not very clear, but vendors claim that implementation within a closed environment are possible. You should check the current status on the regulation.